

IEEE 9-Bus AC N-1 Branch-Outage Analysis

Load Flow Lite

Abstract

This note documents an AC N-1 branch-outage study for the IEEE 9-bus case in `data/case9.json`. Each branch is removed one at a time, the remaining AC power-flow problem is classified topologically, and solvable post-contingency cases are solved with the Newton–Raphson solver in this repository. The checked-in JSON report is `data/case9_n1_results.json`.

1 Scope

N-1 analysis asks whether the system remains acceptable after the loss of one component. Here the components are the nine branch records in the IEEE 9-bus fixture. The study is deliberately narrow:

- AC power flow is solved for connected post-contingency networks and for generator-islanding cases whose surviving component still contains the original slack bus and all load.
- No optimal power flow, economic redispatch, dynamic stability, transient stability, protection action, or load shedding is modeled.
- Branch thermal ratings and generator reactive limits are absent from the fixtures. The report therefore cannot certify thermal-loading or reactive-limit compliance.
- PV-to-PQ conversion is not performed. Solved generator reactive outputs are reported as diagnostics only.

The voltage screening band is $0.95 \leq |V| \leq 1.05$ p.u.. All power-flow solver quantities are on the 100 MVA base unless explicitly shown in MW or MVAR.

2 Topology and Contingencies

The branches split into the six-branch transmission loop and three radial generator-connection branches shown in Table 1. The stored branch orientation is used as the branch identifier.

Table 1: Branch classification used for the case9 N-1 study.

Group	Branches
Transmission loop	4–5, 5–6, 6–7, 7–8, 8–9, 9–4
Generator connection	1–4, 3–6, 8–2

Removing branch 1–4 isolates the slack bus. The remaining load component has no angle reference, so the ordinary AC power-flow problem is not formulated. Removing branch 3–6 or 8–2

isolates a single PV generator bus. In this implementation the surviving slack-containing component is solved, with the missing real generation absorbed by the original slack bus. That is a modeling choice for this small study, not a general graph-theoretic requirement.

3 Base Case

The base case converges in four Newton iterations. Its solved voltage range is 0.9956 p.u. at bus 9 to 1.0400 p.u. at bus 1. The real-power loss is 4.641 MW. The branch reactive endpoint sum,

$$\sum_{\ell} (Q_{\ell, \text{from}} + Q_{\ell, \text{to}}),$$

is -92.160 MVar. This quantity includes the effect of line charging and should not be read as colloquial reactive losses.

4 Solved Contingencies

Table 2 summarizes every solved connected or partial-island case. The largest branch-end flow column is the largest absolute real-power flow at either endpoint of any surviving branch.

Table 2: Solved N-1 results.

Outage	Status	Min $ V $ p.u.	Bus	Loss MW	$Q_f + Q_t$ MVar	Max $ P $ MW
4–5	solved	0.9418	5	6.136	-63.274	163.000
5–6	solved	0.9639	5	9.491	-21.579	163.000
3–6	partial island	1.0028	9	3.629	-88.881	163.000
6–7	solved	0.9783	7	5.353	-62.616	163.000
7–8	solved	0.9690	7	12.093	-28.468	163.000
8–2	partial island	0.9924	9	4.435	-71.541	234.435
8–9	solved	0.9380	9	13.208	0.759	163.000
9–4	solved	0.8388	9	9.567	-38.782	163.000

The largest voltage changes from the base case occur at bus 9 for outage 9–4, where $|V|$ falls by 0.1569 p.u., and at bus 5 for outage 4–5, where $|V|$ falls by 0.0708 p.u.. The 3–6 and 8–2 partial-island cases remain close to the base voltage profile because the transmission loop and all loads remain connected to the original slack bus.

5 Islanding and Voltage Violations

The generator-connection outages are summarized in Table 3.

Table 3: Generator-connection outage treatment.

Outage	Status	Treatment
1–4	slack islanded	Not solved; load component has no slack bus
3–6	partial island	Bus 3 removed; 85 MW absorbed by slack
8–2	partial island	Bus 2 removed; 163 MW absorbed by slack

Table 4: Post-contingency voltage violations for the 0.95–1.05 p.u. band.

Outage	Bus	$ V $ p.u.
4–5	5	0.9418
8–9	9	0.9380
9–4	9	0.8388

Voltage-limit violations are low-voltage violations only:

No solved generator reactive output exceeds the configured diagnostic threshold of 2.0 p.u. on the system base. This is only a soft diagnostic: the fixture has no $Q_{\min}$ or $Q_{\max}$ limits, so no reactive-limit compliance claim is made.

6 Verification

The implementation checks the base case against the existing MATPOWER fixture and checks the six connected transmission-loop outages against the static MATPOWER fixture in `data/case9_n1_solutions.json`. For each solved or partial-island case, active and reactive balance are checked on the surviving component:

$$\sum_i (P_{G,i} - P_{D,i}) = \sum_{\ell} P_{\ell,\text{loss}}, \quad \sum_i (Q_{G,i} - Q_{D,i}) = \sum_{\ell} (Q_{\ell,\text{from}} + Q_{\ell,\text{to}}).$$

The JSON output is deterministic by default and omits timestamps unless the script is called with `-timestamp`.